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# Pedo-hydrological and sediment responses to simulated rainfall on soils of the Konya Uplands (Turkey)

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**Abstract**

Microplot experiments using a portable rainfall simulator were carried out in April 1992 on 15 Turkish soils with textures ranging from clay loam to loamy clay and slopes from 3° to 12°. In order to determine the effects of a change in land use on infiltration and erosion, pairs of sites with soils developed on identical parent material were chosen. Several physical and chemical properties of the soil were determined and related by regression analysis to various soil loss and infiltration variables.

The regression equations showed that plant cover, and therefore the type of land use, was the most important factor. Plant cover influenced various important soil characteristics which determine the infiltration behaviour of the soil, namely humus, root content and macropore distribution. The slope, physical, textural, and chemical properties of the soils and aggregate stability were only of secondary importance. The shear vane test was not well-suited for measuring the surface properties of these soils, which were initially cohesionless and consisted of loose aggregates.

Erosion damage mapping showed high rill erosion rates after snow melting and spring rainfalls on tilled and seedbed sites, up and down hill culture, slopes with gradients > 8 degrees, and on northwest exposed slopes. Results of the microplot tests and erosion damage mapping were in general qualitative agreement. On the catchment scale, the variations of the mesoclimate due to exposure and altitude became important.

## 1. Introduction

Soil loss prediction has been an aim of soil erosion research since the early studies of Middleton (1930). Since then many works have attempted to isolate properties which determine the soil's resistance to erosional processes. The predictive USLE model of Wischmeier and Smith has become a widespread tool for assessing erosion rates, but it

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has limitations when used in different climates, soils or vegetations, or at scales other than the plot scale. The model also provides no information about the processes active in erosion, and so in recent years a great number of experiments, often involving the simulation of rainfall in the laboratory, have been conducted to investigate and model distinct processes such as splash detachment, sheetwash and rilling, soil sealing, etc. (e.g. Al-Durrah and Bradford, 1982; Govers, 1985; Loch and Thomas, 1987; Brunori et al., 1989; Römkens et al., 1990; Cerrera et al., 1991; Merzouk and Blake, 1991).

However, soil characteristics in the field vary widely and have combined effects. Their effects vary also depending on the scale. Experimental field work can measure the combined effects of the various factors and processes, and allows the dominant processes or soil loss determinants for a given soil or land-use form to be determined. Simple-to-measure parameters are required in order to predict infiltration and soil erosion for larger areas and for integration in erosion models. In order to apply plot-measurements to larger areas, e.g. the catchment scale, empirical field measurements at different scales are needed to prove the validity of the micro-scale results (Leser, 1988).

The following study presents results of microplot experiments with a rainfall simulator on soils of a small hydrologic catchment area in south-central Turkey. The objective was to establish the magnitude of the soil erodibility and infiltration variation by means of a comparison of sites representing typical soil forms, slope gradients, and land-use types. Simple-to-measure site characteristics were to be examined with regard to their use in an empirical predictive model and their suitability for describing erodibility and infiltration characteristics.

## 2. The field area

The study area is located in the northern belt of the Taurus Mountains of south-central Turkey, close to Konya and Lake Beysehir (380140 N/320550 E). The climate is subhumid with hot summers and cold winters (annual mean 9–10° C, 490 mm rainfall). The test area is located near the village of Çiftliközü in a smooth sloping area of limestone and conglomeratic rocks with a cover of clayey deposits. The Konya Uplands are an area of dry farming; predominantly wheat is being sown in October. The steeper slopes and poorer soils are not cultivated and support a degraded brush and grass vegetation (Böhm, 1989; Yilmaz, 1991).

## 3. Material and methods

### 3.1. Methods

Fifteen individual test sites representing typical slope gradients, soil types, and land-use forms were selected. In order to determine the direct effects of a change of land use on infiltration, runoff and soil loss, pairs of sites with soils developed of identical parent material were chosen, laying not more than 20 m apart. The paired sites are: (A,

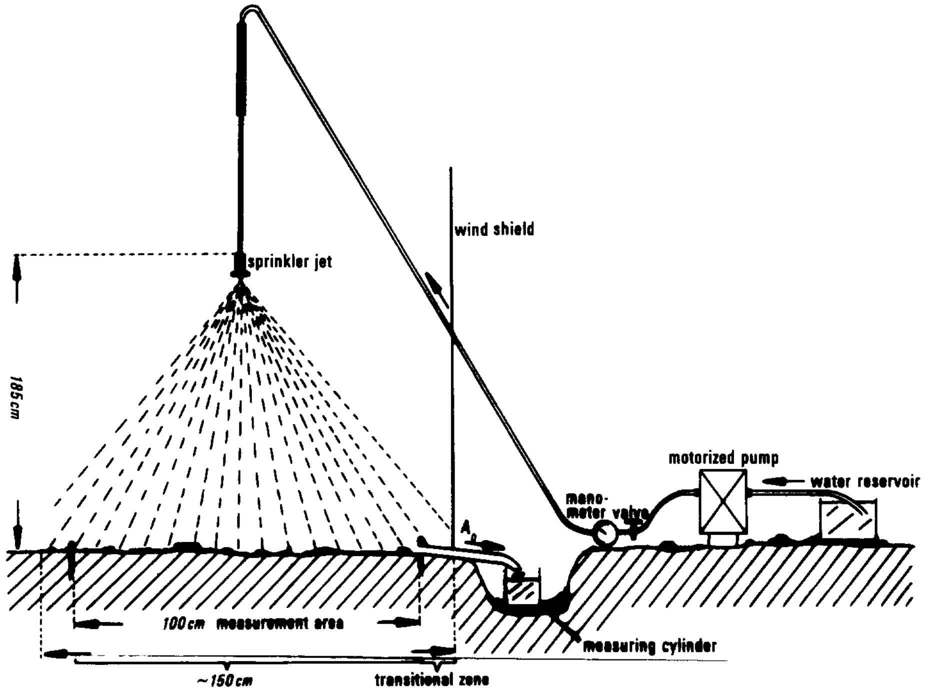


Fig. 1. Schematic view of the rainfall simulator.

B and 9), (C and D), (3 and 4), (13 and 14), (15 and 16) and (19 and 20). Plots 1 and 17 are individual sites. The infiltration and erosion measurements were carried out with a portable rainfall simulator containing a sprinkler jet (Spraying Systems). The design and the main characteristics of the simulator are shown in Fig. 1 and Table 1. Drop characteristics and kinetic energy correspond to rains of an intensity $< 5$ mm/h, whereas the rainfall intensity (120 mm/h) is similar to that of very heavy storms. The soils were tested under field conditions without removing the plant cover or prewetting. Surface runoff was measured volumetrically every minute. Sediment samples were taken every five minutes and were determined gravimetrically in the laboratory. Shear strength

Table 1
Rainfall simulator characteristics

| | |
|---|---|
| Duration of the rainshower | 45 min |
| Mean intensity | 120 mm/h |
| Fall height | 185 cm |
| Mean drop diameter | 0.34 mm |
| Mean velocity | 2.5 m/s |
| Kinetic energy | 4 J/mm |
| Plot size | 1 $m^2$ |
| Sprinkler jet (Spraying Systems) | 1/4 HH 12 SQ |

was measured with an Eijkelkamp pocket vane tester and penetration strength with an Eijkelkamp pocket penetrometer under field moisture conditions with 5 to 10 replicates. Visible macropores (9 classes), root distribution < 2 mm (6 classes), vegetation cover and stone cover were estimated in the field (AG Bodenkunde, 1982). Texture (dry sieving and pipette analysis), stoniness (wet sieving of 1 kg samples), carbonate (Scheibler), organic matter (Walkley and Black), bulk density ($d_B$), cation exchange capacity (CEC) (ammonium acetate extract), exchangeable Na, Ca, Mg (water extract),

Table 2
Soil characteristics

| Plot | Texture | Clay | Silt | Sand | Gravel | Lime | O.M. | Roots < 2 mm | Roots < 2 mm | Macro-pores | Macro-pores |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Dimension: | | % | % | % | % | % | % | rank | rank | rank | rank |
| Depth (cm): | | 0–20 | 0–20 | 0–20 | 0–20 | 0–20 | 0–20 | 0–20 | 20–40 | 0–20 | 20–40 |
| A | scL | 42 | 24 | 34 | 6 | 3 | 1.2 | 4 | 3 | 6 | 1 |
| B | scL | 34 | 26 | 40 | 6 | 7 | 0.8 | 3 | 2 | 2 | 2 |
| 9 | scL | 38 | 23 | 39 | 8 | 4 | 1.2 | 3 | 2 | 3 | 2 |
| C | scL | 43 | 28 | 29 | 8 | 26 | 2.5 | 6 | 4 | 3 | 6 |
| D | scL | 32 | 24 | 44 | 21 | 21 | 5.0 | 6 | 5 | 8 | 6 |
| 1 | C | 72 | 13 | 15 | 1 | 0 | 1.4 | 4 | 3 | 3 | 2 |
| 3 | scL | 39 | 24 | 37 | 8 | 5 | 1.7 | 5 | 4 | 5 | 4 |
| 4 | lC | 47 | 20 | 33 | 16 | 3 | 1.3 | 4 | 2 | 3 | 3 |
| 13 | scL | 33 | 34 | 33 | 8 | 59 | 1.7 | 5 | 3 | 5 | 3 |
| 14 | scL | 33 | 33 | 34 | 6 | 50 | 1.4 | 3 | 3 | 4 | 3 |
| 15 | lC | 46 | 29 | 25 | 12 | 14 | 1.3 | 5 | 3 | 3 | 3 |
| 16 | lC | 51 | 27 | 22 | 5 | 15 | 1.0 | 4 | 3 | 3 | 3 |
| 17 | lC | 51 | 26 | 23 | 10 | 2 | 3.5 | 5 | 5 | 8 | 6 |
| 19 | cL | 30 | 40 | 30 | 12 | 32 | 1.6 | 5 | 4 | 6 | 6 |
| 20 | cL | 36 | 37 | 27 | 5 | 21 | 1.2 | 5 | 3 | 6 | 4 |

| Plot | Shear strength | Shear strength | Penetration strength | $d_B$ | $d_B$ | EC | ESP | Na | CEC |
|---|---|---|---|---|---|---|---|---|---|
| Dimension: | kg/cm$^2$ | kg/cm$^2$ | kg/cm$^2$ | g/cm$^3$ | g/cm$^3$ | mS/cm$^{-1}$ | % | meq/l | meq/100 g |
| Depth (cm): | 5 | 10 | 10 | 0–20 | 20–40 | 0–20 | 0–20 | 0–20 | 0–20 |
| A | 0.30 | 0.8 | 4.0 | 1.39 | 1.50 | 0.65 | 3.5 | 0.8 | 21.5 |
| B | 0.13 | 0.8 | 3.0 | 1.48 | 1.44 | 0.48 | 3.9 | 0.8 | 20.5 |
| 9 | 0.13 | 0.8 | 3.3 | 1.35 | 1.43 | 0.52 | 3.7 | 0.8 | 21.1 |
| C | 0.27 | 0.4 | 1.8 | 1.17 | 1.12 | 0.60 | 1.2 | 0.4 | 32.7 |
| D | 0.55 | 0.4 | 1.8 | 1.08 | 1.34 | 0.65 | 1.2 | 0.6 | 37.6 |
| 1 | 0.33 | 1.1 | 4.3 | 1.15 | 1.24 | 0.54 | 0.9 | 0.3 | 33.4 |
| 3 | 0.77 | 0.4 | 1.5 | 1.36 | 1.36 | 0.60 | 0.9 | 0.3 | 23.2 |
| 4 | 0.27 | 0.8 | 3.6 | 1.38 | 1.45 | 0.60 | 0.9 | 0.3 | 22.4 |
| 13 | 0.49 | 0.7 | 3.0 | 1.30 | 1.34 | 0.71 | 0.4 | 0.1 | 28.5 |
| 14 | 0.19 | 0.6 | 2.5 | 1.28 | 1.35 | 0.77 | 0.4 | 0.1 | 26.3 |
| 15 | 0.15 | 0.5 | 1.3 | 1.32 | 1.24 | 0.71 | 0.2 | 0.1 | 46.0 |
| 16 | 0.13 | 0.5 | 1.0 | 1.30 | 1.35 | 0.71 | 0.3 | 0.1 | 43.2 |
| 17 | 0.38 | 1.1 | 4.3 | 1.24 | 1.29 | 0.48 | 0.6 | 0.3 | 48.2 |
| 19 | 0.13 | 0.5 | 2.3 | 1.31 | 1.41 | 0.77 | 0.9 | 0.2 | 21.6 |
| 20 | 0.15 | 0.4 | 3.5 | 1.35 | 1.40 | 0.65 | 0.4 | 0.1 | 28.8 |

electric conductivity (EC), and antecedent soil moisture content were determined in the laboratory. Aggregate stability (A.S.) was measured as the change in the mean weight diameter ($\Delta$MWD) of prewetted aggregates 3–5 mm in diameter after a 5 minute shower (120 mm/h, 480 J/h). Thirty-two individual site and soil properties were related by stepwise multiple regression analysis to 8 dependent variables describing infiltration and soil loss. The statistical analyses were carried out with the programm SPSS/PC using a confidence interval of 95%. Two runs were computed, one with metric scaled and one with rank scaled data. Soil erosion damage occurring after snow melting and spring rainfalls was mapped in spring 1988, 1992, and 1993 on the basis of the manual of Schmidt (1979) for an area of 10 ha arable land plus 3 ha pasture. Data for the USLE sites in Çiftliközü were taken from Önmez (1991).

### 3.2. Material

The soils of the 15 test sites belong to the types, calcic Cambisols (calcic Ochrept) and Reddish Brown Chestnut Soil (Xeroll). The latter type occurs in two varieties, Reddish Chestnut Soil and calcic Reddish Chestnut soil. The Reddish Chestnut Soils (5YR) have a carbonate content of less than 5%, and contain more than 45% clay. The subsoils generally have a strong compact-blocky structure with a $d_B$ up to 1.65. The aggregates of the surface layer are subangular blocky and those on the soil surface are crumb and loose when dry. Due to erosional processes the topsoils mostly have a reduced clay content and gravel accumulates on the surface (e.g. sites B, D, 3 and 9, Table 2). The calcic Cambisols (7.5YR to 10YR) contain up to 60% silty and sandy carbonates, their clay content is 30% to 40%. Structure is granular to crumb in the topsoil and subangular blocky in the subsoil. Bulk density is moderate ($d_B$ < 1.4).

Table 3
Site characteristics

| Plot | Land use | Plant cover (%) | Slope (deg.) | A.S. ($\Delta$GMD) (mm) | IRAT40 (mm) | POND (mm) | CONC45 (g/l) | SEDALL 45 (g) |
|---|---|---|---|---|---|---|---|---|
| A | pasture | 65 | 9 | 0.5 | 0.90 | 26 | 1.2 | 29 |
| B | seedbed | 10 | 8 | 3.2 | 0.87 | 10 | 6.8 | 138 |
| 9 | fallow | 5 | 9 | 2.4 | 1.23 | 13 | 17.1 | 250 |
| C | seedbed | 20 | 4 | 0.8 | 2.20 | 19 | 1.8 | 45 |
| D | woodland | 100 | 4 | 0.3 | 0.45 | 24 | 0.3 | 2 |
| 1 | pasture | 50 | 3 | 1.7 | 0.45 | 28 | 1.2 | 57 |
| 3 | stubble | 55 | 3 | 1.3 | 1.43 | 47 | 0.8 | 50 |
| 4 | seedbed | 15 | 10 | 1.9 | 0.18 | 15 | 5.7 | 212 |
| 13 | pasture | 90 | 12 | 0.9 | 2.10 | 120 | 0 | 0 |
| 14 | stubble | 20 | 12 | 1.5 | 1.46 | 6 | 1.9 | 29 |
| 15 | stubble | 70 | 4 | – | 1.67 | 37 | 0.7 | 4 |
| 16 | seedbed | 10 | 4 | 2.3 | 1.15 | 11 | 4.1 | 116 |
| 17 | woodland | 70 | 9 | 0.2 | 1.27 | 15 | 0.1 | 5 |
| 19 | pasture | 80 | 4 | 2.5 | 1.96 | 40 | 0.7 | 3 |
| 20 | seedbed | 15 | 4 | 2.9 | 0.88 | 13 | 5.1 | 230 |

IRAT40 = infiltration rate at minute 40, POND = amount of water infiltrated till ponding, CONC45 = sediment concentration in minute 45, SEDALL45 = total sediment discharge after 45 minutes.

Pasture and woodland sites generally have more stable aggregates and contain higher amounts of humus, roots and visible macropores. This is combined with a compact to block-like structure of the soil surface. Due to the disrupted topsoil structure on tilled arable land, soil surfaces contain loose granular to crumb aggregates which are relatively unstable. Shear and penetration strength at depths of 5 and 10 cm are low. Shear strength at the surface could usually not be measured due to the cohesionless state of the crumb aggregates. EC was related to the carbonate content and CEC to the organic matter content.

Surface cover of the woodland sites was regarded as being 100% (site D), because the mineral soil was completely covered by a layer of humus and plant residues. Plant cover is closely correlated to soil organic matter, root distribution, and macropore content ($R_s$ = 0.67–0.69). The slope gradients of the sites tested varied from 3 to 12°.

## 4. Results

### 4.1. Microplot measurements of infiltration and erosion

Due to the low number of tests and the variation of land use on paired sites developed on the same parent material, it is not surprising that the correlation coefficients of the static soil properties, such as texture, are generally low. The best correlations between the infiltration/erosion variables and plot characteristics were computed with plant cover, and so this was included as the first parameter in all regression models (see Tables 4 and 5). As the correlation coefficients in Table 4 indicate, organic matter

Table 4
Correlation coefficients of site/soil-characteristics and infiltration/erosion parameters

| Site characteristics | POND | IRAT40 | CONC45 | SEDALL45 |
|---|---|---|---|---|
| Plant cover | 0.610 | 0.699 | −0.678 | −0.785 |
| Clay 0–20 | | −0.563 | | |
| Clay 20–40 | | −0.565 | | |
| Silt 0–20 | | 0.566 | | |
| Silt 20–40 | | 0.532 | | |
| O.M. % ($R_s$) 0–20 | | 0.514 | −0.620 | −0.521 |
| Bulk density 20–40 | | | 0.493 | |
| Shear strength 5 | | −0.499 | | |
| Penetration strength 10 | | −0.549 | | |
| Lime % 0–20 | 0.528 | 0.638 | | |
| EC 0–20 cm | | 0.539 | | |
| ESP 0–20 | | 0.568 | | |
| Macropores 0–20 ($R_s$) | | 0.475 | −0.614 | |
| Macropores 20–40 ($R_s$) | | 0.541 | −0.475 | −0.531 |
| Roots 0–20 ($R_s$) | | 0.557 | −0.669 | −0.552 |
| Roots 20–40 ($R_s$) | | 0.673 | −0.741 | −0.673 |
| A.S. | | 0.635 | −0.478 | −0.528 |

$n = 15$; confidence interval: $R$ = 5% = 0.4821, 1% = 0.6055; $R_s$: 5% = 0.4429, 1% = 0.6000.

Table 5
Multiple regression equations for the tests

| Parameter | | Multiple $R$ | Variable | Multiple $R^2$ |
|---|---|---|---|---|
| POND | 1. | 0.6096 | plant cover | 0.3716 |
| IRAT40 | 1. | 0.6989 | plant cover | 0.9231 |
| | 2. | 0.866 | penetration strength 10 | |
| | 3. | 0.9391 | lime | |
| | 4. | 0.9608 | % clay 20 | |
| CONC45 | 1. | 0.6778 | plant cover | 0.6279 |
| | 2. | 0.7927 | ESP | |
| SEDALL45 | 1. | 0.7852 | plant cover | 0.7308 |
| | 2. | 0.8549 | $d_B$ 40 | |

Analysis with metric scaled data ($n = 15$).

content, macropores, and roots do significantly influence the infiltration rate as well as the sediment discharge, but due to their strong intercorrelations with the plant cover these variables were not included in the regression equations. A higher clay content and a lower silt content were associated with a decreased infiltration rate. Higher shear and penetration strengths reduced the infiltration rate. The soil strength was heterogeneous, since it was affected by the clay content of the soil, but also modified by the disturbance due to tillage. For instance, with higher clay content penetration strength increased, (sites $1 > A > 13$), but tillage reduced the shear strength of similar sites (e.g. sites $A > B$ and 9). Sites with high carbonate content showed both, high infiltration rates and late ponding ($R = 0.638$ and 0.528, respectively). This may be caused by an increased silt content of the calcic soil forms ($R = 0.782$ for carbonate versus silt), but on the other hand the aggregate stability at these sites is mostly high, and the subangular soil structure might be responsible for the good infiltration. An explanation for the inclusion of the ESP variable might be the intercorrelation with sand at a depth of 20 cm ($R = 0.543$).

The amount of infiltrated water till ponding (POND) was considered to be reached when surface runoff first left the plot. Astonishing is the lack of a statistical relation between POND and the various soil characteristics. Ponding seems to be more sensitive to the dynamics at the soil surface, sealing and aggregate stability. It is also modified by the slope, which can force immediate runoff (see sites 13/14).

The variables describing soil loss were closely related to the organic matter content, which is the factor that influenced aggregate stability most (A.S. versus O.M. — $R = 0.612$, correlation data not presented here).

Plant cover and its related variables dominated the soil loss variables, mainly because of the large reduction in the runoff (e.g. sites 13/14 and 19/20). If comparable runoffs occurred, the cover protected the soil against the rainsplash energy (see sites A/B/9).

The role of the variable bulk density at 20–40 cm depth for the soil loss is not clear. Apart from the infiltration rate it has no obvious causally relation to the processes which take place at the soil surface.

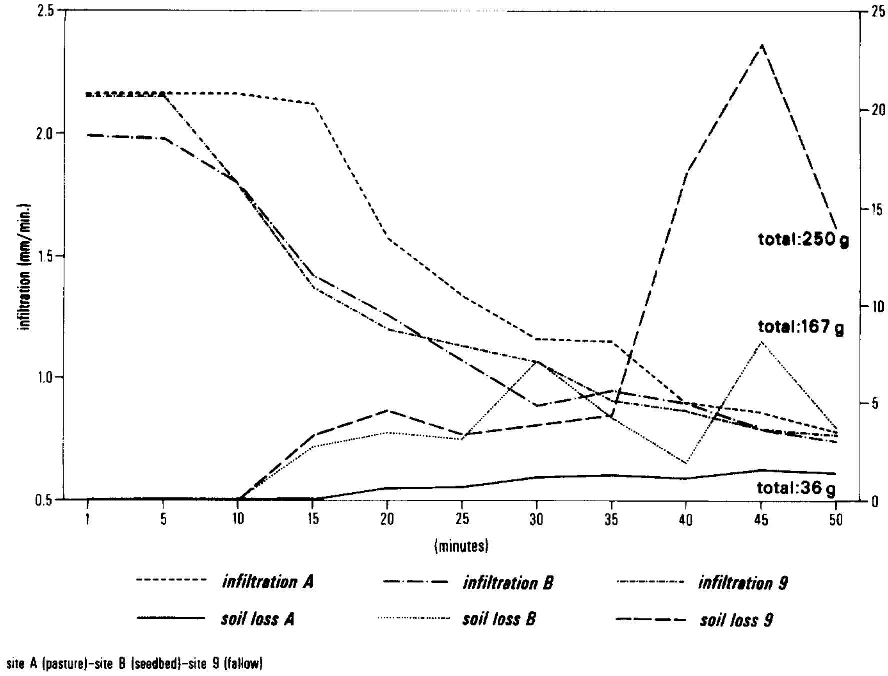


Fig. 2. Reddish Brown Chestnut Soil — infiltration and sediment response to simulated rainfall (sites A, B and 9).

*Examples*

A selection of results illustrating the effect of the land-use form on the infiltration and erosion patterns of identical soils is presented in Figs. 2 and 3. Fig. 2 shows a clayey Reddish Brown Chestnut Soil under pasture, fallow, and seedbed conditions (sites A, 9 and B). Due to the dense subsoil ($d_B = 1.65$) with low amounts of macropores (1–2) hydraulic conductivity on all three sites was low and led to a rapid decrease in infiltration. Equilibrium occurred at final rates of about 0.9 mm/min. On pasture the surface vegetation cover of approximately 65% and the greater aggregate stability reduced the susceptibility to rainsplash. While runoff shear stresses on all three sites were identical (same amounts of runoff water), rainsplash was only sufficient to release aggregates from the surface on the fallow and the seedbed site. There, the sediment suspension caused surface sealing and therefore quick ponding, as indicated by the immediate decrease of the infiltration envelope.

General differences in the erodibility of the three soils are indicated by the aggregate stability test. Stability varied from a ΔMWD of 0.5 cm on the pasture site to ΔMWD 3.2 cm on the seedbed site (see Table 3). However, the soil characteristics tested do not give any information about which of the soil properties were responsible for this difference in aggregate stability. While no visible signs of erosion occurred on the pasture site, on fallow and seedbed the runoff finally incised various small rills of 0.5 to 1 cm depth and 1–2 cm width in which most of the sediment transport took place.

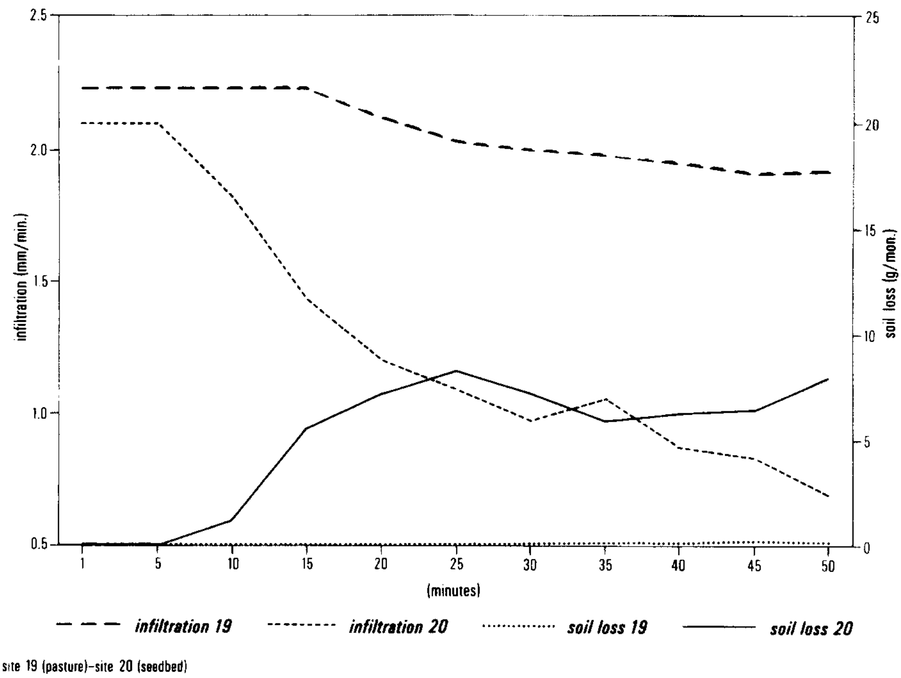


Fig. 3. Calcic Cambisol — infiltration and sediment response to simulated rainfall (sites 19 and 20).

Fig. 3 shows an example of the calcic Cambisol soil form (sites 19 and 20) which is poorer in clay and contains a medium dense subsoil ($d_B = 1.40$) with higher amounts of roots and visible macropores (4–6). Aggregate stability and the shear and penetration strength of the soil at the seedbed site were low and similar to those of the Reddish Chestnut Soil sites described above. By contrast the aggregates at the pasture site with the calcic Cambisol were less stable. Due to the high amounts of macropores and roots, the infiltration capacity on pasture was high. The topsoil of the seedbed plot sealed after 5 minutes. Due to the low aggregate stability ($\Delta$MWD = 2.9 cm) sediment loss reached 230 g in 45 min. Here, too, small rills occurred.

Table 6 shows the results of all 15 tests described above, as well as 7 replications carried out in June. The replications were done after a change of plant cover or land use form. On woodland sites, almost no runoff and no erosion occurred. Pasture, stubble, and wheat sites showed high infiltration rates and similar low to moderate soil losses. Seedbed and fallow sites had short ponding times, low infiltration rates, and huge soil losses. The variation in the values within the classes is probably mainly due to the differences between the soil types and the lower moisture content in June.

## *4.2. USLE plot measurements*

In order to apply the site results at the catchment scale we must check wether they correspond to the erosion patterns occurring in the area. Are the processes or controlling

Table 6
Effects of land use on infiltration and soil loss

| Land use | No. of tests | Infiltration rate (mm/min) | | Ponding time (min) | | Sediment loss (g) | |
|---|---|---|---|---|---|---|---|
| | | mean | min./max. | mean | min./max. | mean | min./max |
| Forest | 3 | 1.9 | 1.3/2.3 | 23 | 7/ > 50 | 3.3 | 0/5 |
| Pasture | 4 | 1.4 | 0.5/2.1 | 25 | 12/ > 50 | 22.3 | 0/57 |
| Fallow | 1 | 0.9 | 0.9 | 6 | 6 | 250 | 250 |
| Stubble | 4 | 1.6 | 1.4/1.8 | 13 | 3/22 | 27.7 | 4/50 |
| Seedbed | 7 | 0.8 | 0.6/1.4 | 7 | 5/9 | 169.1 | 45/273 |
| Wheat > 50% cover [a] | 3 | 1.6 | 0.9/2.7 | 17 | 7/31 | 18.8 | 0/39 |

[a] Measured in June 1992.

factors valid at a larger scale or do other factors become dominant, e.g. slope gradient or slope length? Data from plot measurements, erosion mapping and observations will now be presented to shed light on these questions.

Önmez (1991) presented results of standard USLE plots on Reddish Chestnut Soils in Çiftliközü from a 10 year period. On the pasture plot (our no. 1 plot) no surface runoff and no erosion occurred in all the 10 years. Since the infiltration rate there is according to our findings very low (27 mm/h), it is likely that surface runoff was trapped in cracks and under vegetation, as Imeson et al. (1992) reported for soils in Spain (1992, p. 352). On the KCP plots (similar to our plots B and 9) Önmez obtained values of K = 0.08, C = 0.21 and P = 0.31, which correspond to soil losses of 2.8 t, 0.5 t and 0.2 t/ha/a, respectively. On the C plot 30% of the annual soil loss occurred in a period of 50 days during tillage and seedbed preparation while the soil surface was bare.

### 4.3. Field observations

Erosion damage mapping of a 10 ha area of Reddish Chestnut Soil in March 1988, 1992 and 1993 showed mean soil losses of 2 t/ha per year. Ninety-three percent of the losses occurred on seedbed plots (3.4 t/ha per year), the remaining on stubble and fallow (Table 7). The soil loss was 4.9 t/ha in 1988 and < 0.6 t/ha in 1992 and 1993. The maximum soil loss on a single field was 84 t/ha (seedbed). An area of 3 ha pasture showed no visible linear erosion. The main damage occurred on seedbed sites, sites with up and down hill culture, on slopes with gradients > 8°, and on northwest exposed slopes.

Observations in March 1992 showed severe erosion damage on agricultural sites due to snow melting on the flat, north-facing open areas of the higher, northern parts of Çiftliközü. Flooding with water up to 40 cm deep occurred because the meltwater could not infiltrate into the frozen ground. By contrast, on the lower parts and south facing slopes, which are exposed to higher levels of solar radiation, the snow had mostly melted away before the main melt started and no erosion occurred. Since the topsoils of the north facing areas remained wet for a long time, they were more easily eroded by the

Table 7
Erosion forms and soil loss on 10 ha arable land and 3 ha pasture according to erosion damage mapping

| | | Rins (No.) | Rills (No.) | Soil loss | | |
|---|---|---|---|---|---|---|
| | | | | (t) | (t/form) | (t/ha) |
| Slope (deg.) | 2–4 | 4 | 14 | 6.8 | 0.38 | |
| | 5–8 | 4 | 22 | 13.0 | 0.50 | |
| | 9–12 | 3 | 32 | 40.3 | 1.15 | |
| Exposure | SW | 0 | 2 | 0.3 | 0.14 | |
| | SE | 1 | 24 | 12.2 | 0.49 | |
| | NW | 10 | 42 | 47.6 | 0.92 | |
| Land use | seedbed | 10 | 59 | 55.7 | 0.81 | 3.4 |
| | stubble | 1 | 1 | 0.9 | 0.45 | 0.1 |
| | rough fallow | 0 | 8 | 3.5 | 0.44 | 0.6 |
| | pasture | 0 | 0 | 0.0 | 0.00 | 0.0 |
| Direction | up and down | 10 | 51 | 53.9 | 0.88 | 2.5 |
| | contour | 1 | 17 | 6.2 | 0.32 | 0.7 |

spring rainfalls. This higher erosivity may be one reason for the higher erosion rates of the north facing slopes.

The special case of extreme runoff and soil loss on an almost flat area is a contradiction to the assumption that runoff and erosion are normally related to the gradient. For snow melt events it might be the contrary case in the investigation area. Nevertheless, it was observed that on a larger scale, microclimatic field conditions determined by the relief properties exposure and altitude can have a great importance for the actual erosion events in the catchment.

## 5. Discussion

The objective of the study was to find the magnitude of infiltration and soil loss variation in a catchment area and their controlling factors. It seems out of scope to use the presented data for a comparison with results of experimental laboratory research since in our work only one factor, the rainfall intensity, was controlled while the site properties plant cover and land use showed a big magnitude of variation.

The results have some limitations because of the lack of replications and the use of low energy rainfall, which may underestimate soil loss for a given runoff. Nevertheless, the drop characteristics produced are comparable with natural rainfall of low intensities, which are typical for more than 90% of the amount of rainfall in the study area.

The regression equations illustrated the dominant role of the plant cover. This fact is well known and considered in erosion prediction (Wischmeier, 1960). Plant cover decreases the rainsplash energy (Al-Durrah and Bradford, 1982; Römkens et al., 1990) and therefore soil sealing, reduces the incision of rills (Govers, 1991) and promotes infiltration (Hills, 1971). Our data show that plant cover controls at the same time various important soil characteristics which are controlling factors for the runoff generation as a secondary erosive agence. This properties are namely macropore distribution, soil surface structure, humus content and root content. The latter two variables found Lavee et al. (1991) to be the controlling factors of infiltration along a transect in Israel. In our data they all are closely related to the surface cover and therefore the land use form, whereas slope, textural, soil physical and chemical properties, which also do have an influence on e.g. structural state and aggregate stability, are independent properties with less influence. The combined effects of the canopy cover were more powerful than any other single variable showing dominance in other works, e.g. aggregate stability and shear strength (Coote et al., 1988) or soil moisture (Cerrera et al., 1991). Shear and penetration strength in our data reflected textural properties (clay content) and were modified by the structural state of the topsoil due to the tillage effects. Soil shear strength, proposed by Brunori et al. (1989) as a good measure of soil resistance to erosion showed only poor correlation. Shear vane test was not well-suited to describe surface conditions of initial cohesionless bare tilled soils with loose aggregates. In contradiction to the findings of Merzouk and Blake (1991, p. 546) soils with active $CaCO_3$ showed good aggregation and stable aggregates. The most powerful variable in the regression models, the vegetation cover term, can be used as a predictor for the relative ranking of infiltration, runoff generation and interrill erodibility in the study area.

The incision of rills in loose aggregates indicates that soil loss on microplots is not exclusively a result of rainsplash, as Merzouk and Blake (1991) propose. Our observations showed, that even short flow distances along with a discharge of 1.2 l/min can generate microrills. However, the energy was not sufficient to incise rills in a coherent soil matrix. Discharges of > 3 l/s on clayey soils (Loch and Thomas, 1987) or shear velocities greater 3.0 cm/s (Govers, 1985) are needed.

The results and controlling factors of the microplot tests seem to be in general aggreement with findings of the larger scaled works. All together showed only slight or no measurable erosion and runoff on a pasture site. Likewise the microplot measurement results with severe rainsplash and sheet erosion on arable sites, high rill erosion rates occurred on seedbed sites after snow melting and spring rainfalls. While slope gradient had only a modifying effect on the microplots, it gains importance in the field scale with greater slope length. Slopes of gradients more than 8 degrees showed significantly higher rill erosion rates. On the catchment scale, the variations of the mesoclimate due to exposure and altitude were able to change slope and cover influences in the special case of a snowmelt event. On the contrary, floodings of snowmelt water appeared on almost flat areas. A similar effect caused by persistent rainfall in coincidence with frozen topsoil was described by Gerold and Molde (1989) for the Wendebach catchment, Germany.

## 6. Conclusions

The results show the important role of the plant cover for the infiltration and erosion dynamics of the soils tested. Plant cover not only prevents rainsplash but also provides high infiltration capacities due to its positive effects on humus, root and macropore content and soil structure. In the investigation area, the Karabalçik catchment, arable land must be responsible for the direct runoff and sediment discharge, whereas soils on forests and pasture sites control the water balance of the watershed.

The low intensity rainfall simulation tests on microplots described here were well-suited for determining relative differences in infiltration, rainsplash and sheet erosion between sites. Results from different scales show, that the overall pattern of erosion in the area cannot be adequately understood using only plot and microplot measurements, and so supplementary field observations and mapping of the processes and events currently occurring must be undertaken.

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## References

Al-Durrah, M. and Bradford, J.M., 1982. The mechanism of raindrop splash on soil surfaces. Soil Sci. Soc. Am. J., 46: 1086–1092.

AG Bodenkunde, 1982. Bodenkundliche Kartieranleitung. Hannover, 331 pp.

Böhm, P., 1989. Untersuchungen zum Einfluß der Bodenerosion auf den Bodenwasserhaushalt und die Ertragsfähigkeit landwirtschaftlich genutzter Flächen in Ciftliközü, Provinz Konya (Türkei). Thesis, Geographisches Institut der Universität Göttingen. Göttingen, 133 pp.

Brunori, F., Penzo, M.C. and Torri, D., 1989. Soil shear strength. Its measurement and soil detachability. Catena, 16: 59–71.

Cerrera, M., Clotet, N. and Guardia, R., 1991. Response to rainfall simulation from scarcely vegetated and non-vegetated Badlands. Catena Suppl., 19: 39–56.

Coote, D.R., Malcom-McGovern, C.A., Wall, G.J., Dickinson, W.T. and Rudra, R.P., 1988. Seasonal variation of erodibility indices based on shear strength and aggregate stability in some Ontario Soils. Can. J. Soil Sci., 68: 405–416.

Gerold, G. and Molde, P., 1989. Einfluß der Pedo-hydrologischen Einzugsgebietsvarianz auf Oberflächenabfluß und Stoffaustrag im Einzugsgebiet des Wendebaches. Göttinger Geogr. Abh., 86: 81–94.

Govers, G., 1985. Selectivity and transport capacity of thin flow in relation to rill erosion. Catena, 12: 35–49.

Govers, G., 1991. A field study on topographical and topsoil effects on runoff generation. Catena, 18: 91–111.

Hills, R.C., 1971. The influence of land management and soil characteristics on infiltration and the occurrence of overland flow. J. Hydrol., 13: 163–181.

Imeson, A.C., Verstraten, J.M., Van Mulligen, E.J. and Sevink, J., 1992. The effects of fire and water repellency on infiltration and runoff under mediterranean type forest. Catena, 19: 345–361.

Lavee, H., Imeson, A.C., Pariente, S. and Benyamini, Y., 1991. The response of soils to simulated rainfall along a climatological gradient in an arid and semiarid region. Catena Suppl., 19: 19–37.

Leser, H., 1988. Methodische Probleme Regionaler Bodenerosionsforschungen. Geomethodica, 10: I–VIII.

Loch, R.J. and Thomas, E.C., 1987. Resistance to rill erosion. Observations on the efficiency of rill erosion on a tilled clay soil under simulated rain and run-on water. Catena Suppl., 8: 71–83.

Merzouk, A. and Blake, G.R., 1991. Indices for the estimation of interrill erodibility of Moroccan Soils. Catena, 18: 537–550.

Middleton, H.E., 1930. Properties of soil which influence soil erosion. USDA Techn. Bull., 178.

Önmez, O., 1991. Konya Beysehir sartlarinda üniversal toprak kaybi denkleminin R, K, C ve P faktörleri. Konya Köy Hizmetleri Arastirma Enstitüsü Müdürlügü Yayinlari, 147: 47 pp. [USLE-Factors for Konya-Beysehir].

Römkens, M.J.M., Prasad, S.N. and Parlange, J.Y., 1990. Surface seal developement in relation to rainstorm intensity. Catena Suppl., 17: 1–11.

Schmidt, R.G., 1979. Probleme der Erfassung und Quantifizierung von Ausmaß und Prozessen der aktuellen Bodenerosion auf Ackerflächen. Physiogeographica, 1: 240 pp.

Wischmeier, W.H., 1960. Cropping management factor evaluations for a universal soil-loss equation. Soil Sc. Soc. Am. Proc., 24: 322–326.

Yilmaz, A., 1991. Konya–Çiftliközü, Karabalçik Deresi Havzasi Akimlari (Ara raporu 1979–1988) [Runoff of the Konya–Çiftliközü–Karabalçik Creek, 1979–1988)]. Konya Köy Hizmetleri Arastirma Enstitüsü Müdürlügü Yayinlari, 141: 76 pp.